INCH-POUND

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SUPERSEDING
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21 June 1996

DETAIL SPECIFICATION

ROCKET MOTOR, MK 117 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Mk 117 Mod 0 Rocket Motor for the BQM-74C target drone.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to Thomas.Omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
DOD-STD-2101	-	Classification of Characteristics

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (CAGE CODE 30003) DRAWING

DL1638AS120	-	Rocket Motor, Mk 117 Mod 0

(Copies of this drawing are available from the Commander, Indian Head Division, Naval Surface Warfare Center, Code 840M, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ASQC-Z1.4	-	Procedures, Sampling and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E1742	-	Radiographic Examination (DoD adopted)

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(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance to documents. The rocket motor shall conform to this specification and the documents listed in DL1638AS120.

3.3 Characteristics.

3.3.1 Radiographic inspection (M101). All rocket motor assemblies shall be radiographically inspected (see 4.5.2).

3.3.2 Circuit resistance (M102). The igniter circuit resistance on all rocket motor assemblies, when measured from the connector at 70±10°F, shall be between 0.45 to 0.75 ohms (see 4.5.3).

3.4 Ballistic requirements (M103). Rocket motors shall meet the ballistic requirements in Table I over the temperature range of -40°F to 140°F (see 4.5.5, 6.5.1, 6.5.3, 6.5.4, 6.5.5, 6.5.6).

TABLE I. Ballistics requirements.

Parameter	Requirement
Total impulse (lb f-sec)	3400 min
Action time (sec)	2.5 max
Thrust (lb force)	3750 max
Ignition delay (sec)	0.085 max

3.5 Workmanship. Motors and components, when examined visually in 4.5.1, shall be free from dents, scratches, split or cracked components, poor quality welds, chipping or flaking of plated or painted surfaces, sharp edges, or other defects or discontinuities as specified on the applicable drawings.

3.6 Data cards. Data cards, when completed in accordance with MIL-STD-1168, shall be included for each production lot of rocket motors.

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3.7 Lot numbering. Lot numbering shall be in accordance with MIL-STD-1168.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the applicable test methods of 4.5.

4.3 First article inspection. The first article sample shall consist of 18 rocket motors. The sample shall undergo the tests and inspections specified in Table II. Each rocket motor shall be visually inspected and undergo circuit resistance testing after each test specified in Table II. Failure of any rocket motor shall be cause for first article rejection.

4.4 Conformance inspection. Conformance inspection shall include the tests and inspections in Table III. Failure of any rocket motor shall be cause for rejection.

4.4.1 Lot formation. A lot shall consist of all the rocket motor units loaded from the same propellant grain lot and the same igniter lot and submitted for inspection and test at the same time.

4.4.2 Sampling.

4.4.2.1 Nondestructive tests. Unless otherwise specified, all motors shall be nondestructively inspected in accordance with Table III. There shall be no defects.

4.4.2.2 Destructive test. Destructive tests shall be the ballistic tests in accordance with Table III. The sampling shall be in accordance with ASQC-Z1.4, normal sampling plan, inspection level S4, for each temperature tested. The destructive test shall be evaluated and accepted in accordance with ASQC-Z1.4, standard deviation method, single or double specification limit (as applicable), variability unknown.

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TABLE II. First article inspection.

Treatment or test	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Test paragraph
Visual	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	4.5.1
X-ray and circuit resistance	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	4.5.2, 4.5.3
High temperature storage									x	x	x	x	x	x	x	x	x	x	4.5.4.1
X-ray and circuit resistance									x	x	x	x	x	x	x	x	x	x	4.5.2, 4.5.3
Temperature shock											x	x	x	x	x	x	x	x	4.5.4.2
X-ray and circuit resistance											x	x	x	x	x	x	x	x	4.5.2, 4.5.3
Salt fog											x	x							4.5.4.3
Vibration																			4.5.4.3
@ -40° F											x				x		x		
@ 140° F												x				x		x	
X-ray and circuit resistance											x	x			x	x	x	x	4.5.2, 4.5.3
Two-foot drop											x	x							4.5.4.5
X-ray and circuit resistance											x	x							4.5.2, 4.5.3
Acceleration firings							x	x											4.5.4.6
Ballistic test																			4.5.5
@ -40° F	x	x	x						x		x		x		x	x			
@ 140° F				x	x	x				x		x		x			x	x	

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements not annotated with a classification code are classified minor.

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TABLE III. Conformance inspection.

Inspection or test	Requirement paragraph	Test paragraph
Nondestructive inspection		
Visual	3.5	4.5.1
Radiographic	3.3.1	4.5.2
Circuit resistance	3.3.2	4.5.3
Destructive inspection		
Ballistic test	3.4	4.5.5
@ -40°F		
@ 140°F		

4.5 Test methods

4.5.1 Visual. Motors and components shall be visually inspected to ensure the requirements of 3.5 are met.

4.5.2 Radiographic inspection (M101). Each rocket motor assembly shall be subjected to radiographic inspection in accordance with ASTM-E1742, radiographic quality 2-1T for conformance to DL1638AS120 (see 3.3.1).

4.5.3 Circuit resistance (M102). The rocket motor igniters shall be tested for circuit resistance by applying 0.010 amperes (maximum) to the initiator assembly circuit at 70±10°F.

4.5.4 Environmental and operational tests.

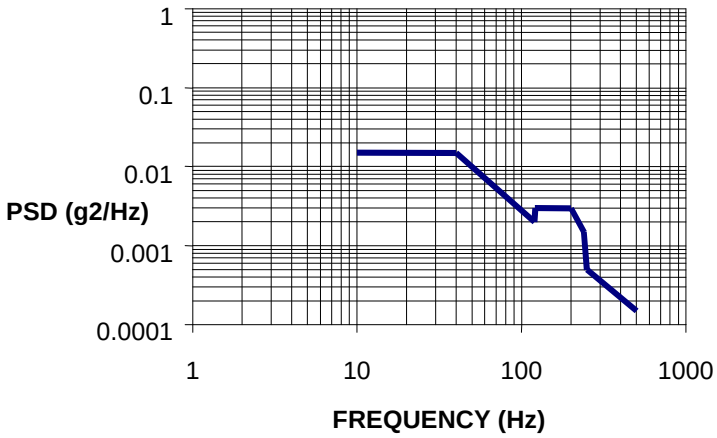
4.5.4.1 High temperature storage. Rocket motors shall be placed in a conditioning chamber at 140° F for 45 days in accordance with MIL-STD-810, Method 501.4, Procedure I.

4.5.4.2 Temperature shock. The rocket motors shall withstand temperature shock in accordance with MIL-STD-810, Method 503.4, Procedure I, Steady State. The motors shall be equally divided into two groups with each group subjected to alternate conditioning chambers at -40°F and 140°F. One group shall begin and end at -40°F, the other at 140°F. The motors shall remain in each chamber for not greater than 8 hours, then transferred within 5 minutes to the other chamber. Each motor shall undergo a total of 3 cycles (see 6.5.2). After temperature cycling, the grain shall be returned to ambient temperature and radiographically inspected as specified in 4.5.2.

4.5.4.3 Salt fog. The rocket motor shall be subjected to an aqueous salt atmosphere in accordance with MIL-STD-810, Method 509.4, Procedure I, Aggravated Screening. Test duration shall be 48 hours. After exposure, the motor shall be allowed to dry for 48 hours under ambient conditions.

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4.5.4.4 Transportation vibration. The rocket motors shall be subjected to a composite version of MIL-STD-810, Method 514.5, Procedure I, Category 4, Truck Transportation over U.S. Highways, for 3 hours per axis for a total of 9 hours. The composite transportation vibration profile is provided in Figure 1. The rocket motors shall be in their shipping configuration for the test. The rocket motors shall be temperature conditioned to the temperatures specified in Table II for 8 to 9 hours prior to test and maintained at the temperature specified in Table II for the duration of the test.



Breakpoints

Freq/PSD

10/0.01500	40/0.01500	120/0.00200	121/0.00300
200/0.00300	240/0.00150	250/0.00050	500/0.00015

FIGURE 1. Transportation vibration profile for vertical, longitudinal, and transverse axes.

4.5.4.5 Two-foot drop. The uncrated rocket motors shall withstand a drop of 2 feet measured from the lowest point on the motor to the impact surface. The impact surface shall be a 2-inch thick steel plate (Brinell hardness of not less than 207) embedded in not greater than 2 feet of reinforced concrete. Motors shall be dropped horizontally. Motors shall be temperature conditioned for not greater than 8 hours at -40°F before testing. Motors shall be drop tested within 15 minutes after removal from conditioning. If out of conditioning for more than 15 minutes, the motors shall be reconditioned at -40°F. Reconditioning time shall be 2 minutes for each minute (over 15) the motor was out of the conditioning chamber. Reconditioning time shall be not greater than 8 hours.

4.5.4.6 Acceleration. During firings, the rocket motor shall withstand a simulated acceleration of 15 g's through the longitudinal axis in accordance with MIL-STD-810, Method 513.4.

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4.5.5 Ballistic test (M103). Rocket motors shall be ballistically tested in accordance with 4.5.5.1 and 4.5.5.2 (see 3.4).

4.5.5.1 Test setup. The rocket motor shall be mounted on a test stand, equipped with a recording oscillograph and instrumentation for recording thrust and time. The motor shall be fired not greater than 15 minutes after removal from the conditioning chamber. If this time limit is exceeded, the motors shall be reconditioned 2 minutes for every minute over 15 that the out of specification condition exists, or not greater than 8 hours. Motors shall be fired by energizing the igniter circuit with 28 volts dc at 7 +0.5, -0.0 amperes.

4.5.5.2 Instrumentation. All measurements shall be made with instruments and systems with a total system accuracy, reproducibility, and frequency response that has been approved by the procuring activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motors provide auxiliary thrust to launch, from either ground or ship, the BQM-74C target drone. The rocket motors were developed specifically for this purpose and have no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Mark and Mod of the rocket motor.
c. Inspection conditions, if other than as specified (see 4.2).
d. Whether first article is required (see 4.3 and 6.3).
e. Packaging (see 5.1 and 6.4).

f. That the safety precaution requirements of the “Contractor’s Safety Manual for

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Ammunition, Explosives, and Related Dangerous Material," DOD-4145.26M are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 18 rocket motors. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Information pertaining to first article may be obtained from Commander, Indian Head Division, Naval Surface Warfare Center, Code 840M, 101 Strauss Avenue, Indian Head, MD 20640-5035.

6.4 Packaging. Contracting officers are advised that the preferred packaging instructions for the Mk 117 Mod 0 rocket motor are contained in Drawing 1638AS123, "Packing and Marking, Rocket Motor, Mk 117 Mod 0."

6.5 Definitions.

6.5.1 Action time. Action time is the time between the initial 200-pound thrust level and the final 200-pound thrust level (see 3.4).

6.5.2 Cycle. A cycle is defined as -40°F to 140°F and back to -40°F, or alternatively from 140°F to -40°F and back to 140°F (see 4.5.4.2).

6.5.3 Ignition delay. Ignition delay is the time from zero time (moment current is applied to igniter) to the initial 200-pound thrust (see 3.4).

6.5.4 Maximum thrust. The maximum thrust is the highest thrust attained by the motor, after 100ms, during a static-firing test, recorded to the nearest lb force (see 3.4).

6.5.5 Thrust-time curve for rocket motor, Mk 117 Mod 0 (see 3.4 and figure 2).

6.5.6 Total impulse. Total impulse is the integral of thrust over the action time (see 3.4).

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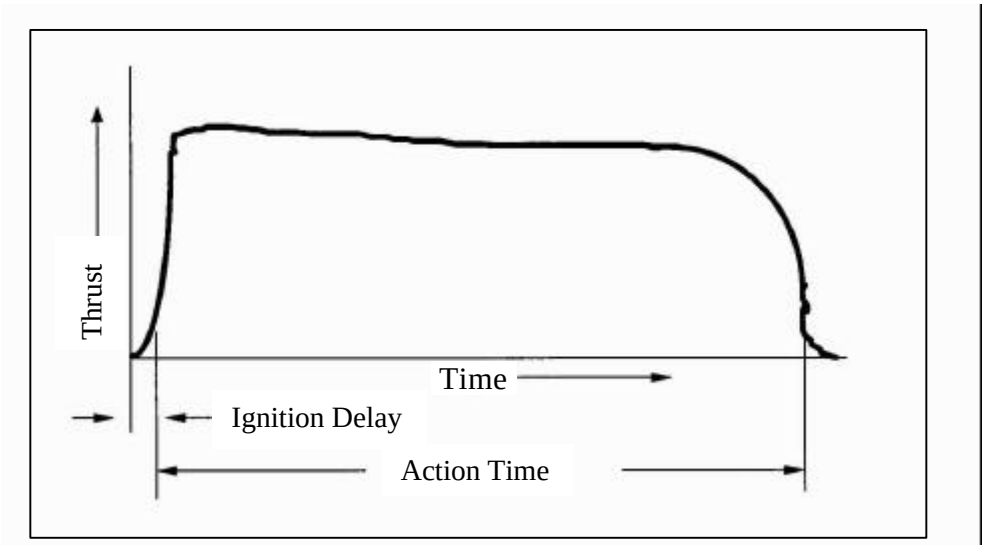


FIGURE 2. Thrust-time Curve

6.6 Subject term (key word) listing.

Ignition delay
Thrust
Total impulse

6.7 Changes from the previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous issue.

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